

RSA Algorithm

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Reference

- Computer Security: Principles and Practice
- National Institute of Standards and Technology
- OWASP

RSA Algorithm

1. Introduction to RSA Algorithm

The RSA Algorithm is one of the most widely used **asymmetric encryption algorithms** in computer security.

RSA is named after its inventors:

- Ron Rivest
- Adi Shamir
- Leonard Adleman

RSA uses **two different keys**:

1. **Public Key** → Shared with everyone
2. **Private Key** → Kept secret

2. Why RSA is Needed

In symmetric encryption, both sender and receiver use the same key.

Problem:

- How can both parties securely share the secret key?

RSA solves this problem by allowing:

- Safe communication over the internet
- Secure key exchange
- Digital signatures

3. Easy Real-Life Analogy of RSA

Imagine a **locked mailbox**:

- The mailbox is open for everyone.
- Anyone can drop a letter inside.
- But only the owner has the key to open it.

In RSA:

- Public Key = Open mailbox
- Private Key = Secret key of mailbox

Anyone can encrypt data using the public key, but only the owner can decrypt it using the private key.

4. Basic Working of RSA

RSA works in three main steps:

Step 1: Key Generation

The receiver creates:

- Public Key
- Private Key

Step 2: Encryption

Sender encrypts message using the public key.

Step 3: Decryption

Receiver decrypts message using the private key.

5. RSA Mathematical Idea

RSA is based on:

- Prime numbers
- Modular arithmetic
- Difficult factorization problem

The security of RSA depends on the difficulty of factoring very large numbers.

6. RSA Key Generation Process

Step 1: Select Two Prime Numbers

Example:

```
[  
p = 3, q = 11  
]
```

Step 2: Calculate n

```
[  
n = p * q  
]
```

```
[  
n = 3 * 11 = 33  
]
```

Step 3: Calculate Euler's Totient Function

```
[  
 $\phi(n) = (p-1)(q-1)$   
]
```

$\phi(n) = (p-1)(q-1)$

```
[  
 $\phi(n) = (3-1)(11-1) = 20$   
]
```

Step 4: Choose Public Key e

Choose:

```
[  
e = 3  
]
```

Condition:

```
[  
 $\gcd(e, \phi(n)) = 1$   
]
```

Step 5: Calculate Private Key d

$$\begin{bmatrix} d * e \bmod \phi(n) = 1 \end{bmatrix}$$

$$d * e \bmod \phi(n) = 1$$

Here:

$$\begin{bmatrix} d = 7 \end{bmatrix}$$

7. RSA Encryption Formula

Encryption formula:

$$C = M^e \bmod n$$

Where:

- (C) = Ciphertext
- (M) = Original message
- (e) = Public key
- (n) = Modulus

8. RSA Decryption Formula

Decryption formula:

$$M = C^d \bmod n$$

Where:

- (M) = Original message
- (C) = Ciphertext
- (d) = Private key

9. Simple RSA Example

Suppose:

- Message = 4

- Public key = (3,33)

Encryption:

$$\begin{bmatrix} C = 4^3 \bmod 33 \end{bmatrix}$$

$$\begin{bmatrix} C = 64 \bmod 33 \end{bmatrix}$$

$$\begin{bmatrix} C = 31 \end{bmatrix}$$

Ciphertext becomes:

$$\begin{bmatrix} 31 \end{bmatrix}$$

Now decrypt:

$$\begin{bmatrix} M = 31^7 \bmod 33 \end{bmatrix}$$

Original message becomes:

$$\begin{bmatrix} 4 \end{bmatrix}$$

10. Applications of RSA

RSA is used in:

- Secure websites (HTTPS)
- Online banking
- Secure email
- VPNs
- Digital signatures
- SSL/TLS certificates

11. Advantages of RSA

- Secure communication
- No need to share secret key
- Supports digital signatures
- Widely trusted

12. Disadvantages of RSA

- Slower than symmetric encryption
- Requires large key sizes
- Complex mathematical operations

13. Scenario-Based Example

Scenario

Ali wants to send his ATM PIN securely to the bank.

Process

1. Bank shares public key.
2. Ali encrypts PIN using public key.
3. Hacker intercepts message but cannot read it.
4. Bank decrypts using private key.

Result:

- Secure communication achieved.

14. Exercise Questions

Short Questions

1. What is RSA?
2. Define public key and private key.
3. Why are prime numbers important in RSA?

4. What is modular arithmetic?
5. What is the role of (e) in RSA?

Long Questions

1. Explain RSA key generation process.
2. Differentiate symmetric and asymmetric encryption.
3. Explain RSA encryption and decryption with example.